

MTH313M PRACTICE PROBLEMS WEEK 1

Question 1. Let $\{N_t\}_{t \geq 0}$ be a Poisson Process with rate λ .

(a) Compute $\mathbb{E}[N_t N_{t+h}]$ for any $t \geq 0, h > 0$.

(b) Let $\{\tilde{N}_t\}_{t \geq 0}$ be another Poisson Process with rate $\tilde{\lambda}$, independent of $\{N_t\}_{t \geq 0}$. Show that $\{N_t + \tilde{N}_t\}_{t \geq 0}$ is a Poisson Process with rate $\lambda + \tilde{\lambda}$.

Question 2. Let $T_1, T_2, \dots$ be i.i.d. RVs with the common distribution χ^2 with 2 degrees of freedom. Consider the counting process defined by

$$N_t := \max\{n : T_1 + T_2 + \dots + T_n \leq t\}, \forall t \geq 0.$$

Find the p.m.f. of N_t for each $t \geq 0$.

Question 3. Complete the exercises left out in class.